



— ESPRIT · FINAL YEAR PROJECT · 2026



AI-Powered Recruitment Platform

From job discovery to evidence-based hiring decisions.

BIBANI Mohamed Habib Allah

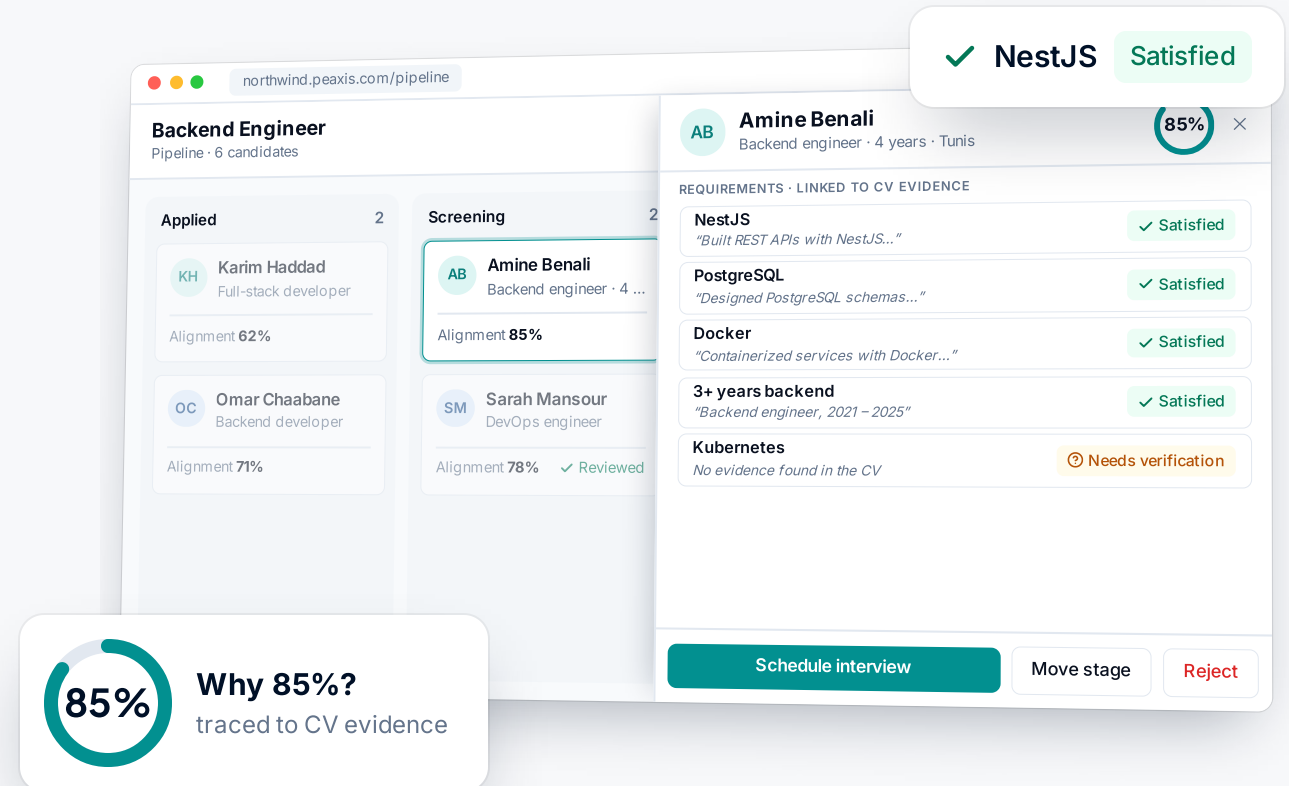
École Supérieure Privée d'Ingénierie et de Technologie — ESPRIT

ACADEMIC SUPERVISOR

Mme Olfa Mannai

COMPANY SUPERVISOR

Mr. Fedi Naimi





— A SHORT STORY · 1 OF 4



Amine

Backend engineer, looking for a new opportunity

✓ CV submitted

A JOB OFFER

Backend Engineer

Northwind · Tunis · Hybrid

NestJS

PostgreSQL

Docker

✓ It seems to match his experience

cv_amine_benali.pdf



...and waits...



— A SHORT STORY · 2 OF 4



The recruiter

Applications keep coming in

AB

Amine Benali
Backend engineer · Tunis

85%

match

Should Amine move forward?

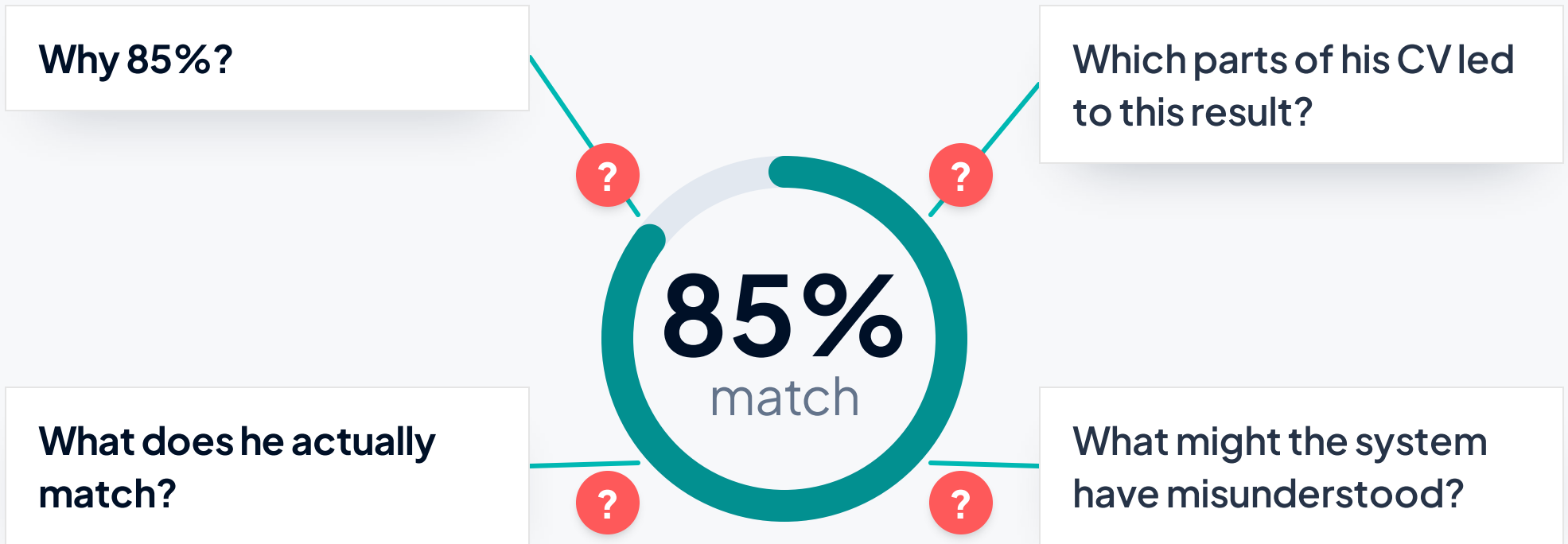
Invite to interview

Reject



— A SHORT STORY · 3 OF 4

Before deciding, several questions remain





— A SHORT STORY · 4 OF 4

The real question is not whether the result is ~~85%~~.

Can we trust a number if we don't understand how we got there?



— INTRODUCTION

Presentation plan

From the recruitment problem to a working product.

01	Context & problem	Why recruitment needs clearer decisions
02	Methodology & requirements	How I organized and defined the project
03	The PEAXIS platform	The candidate and recruiter journey
04	Architecture, AI & production	How the system works and stays reliable
05	Results, perspectives & live demo	What exists today and what comes next



— CONTEXT

Built at **Prospecter**, an AI product company

THE HOST COMPANY

AI product for B2B prospecting, based in Doha, Qatar.

I worked there full time as a software engineer: AI, software, and a real product in production.

MY WORK COVERED THREE AREAS



The application

Web platform, API, business rules



The AI

LLM integration, evidence assessment



The infrastructure

Deployment, security, operations



PROBLEM

Hiring, seen from both sides



CANDIDATE

Sends a CV, then waits



RECRUITER

Receives applications



Reads



Compares profiles



Organizes interviews



When applications grow, the work becomes hard to manage



≈ **1.5 months**

median time to fill a role (SHRM)



37%

of recruiting teams already try generative AI (LinkedIn)



— PROBLEM

Three **problems** to solve

01



Manual work

Reading and comparing CVs takes a lot of time.

02



Scattered tools

The CV is in one place, the discussion in email, the interview in a calendar.

03



Unclear AI results

A score appears, but the recruiter cannot see why.



PROBLEM

Existing solutions, and our difference

TOOLS COMPANIES USE TODAY

LinkedIn Recruiter Sourcing candidates

Greenhouse Structured hiring

Ashby All-in-one suite

Workable Easy for small teams

PEAXIS

Evidence behind every AI result

One journey, from candidate to recruiter
A free plan, with local billing



PROJECT OBJECTIVE

How can AI make recruitment more efficient, while keeping decisions understandable?



Bring the workflow
together



Understand the
candidate's
information



Assist the recruiter

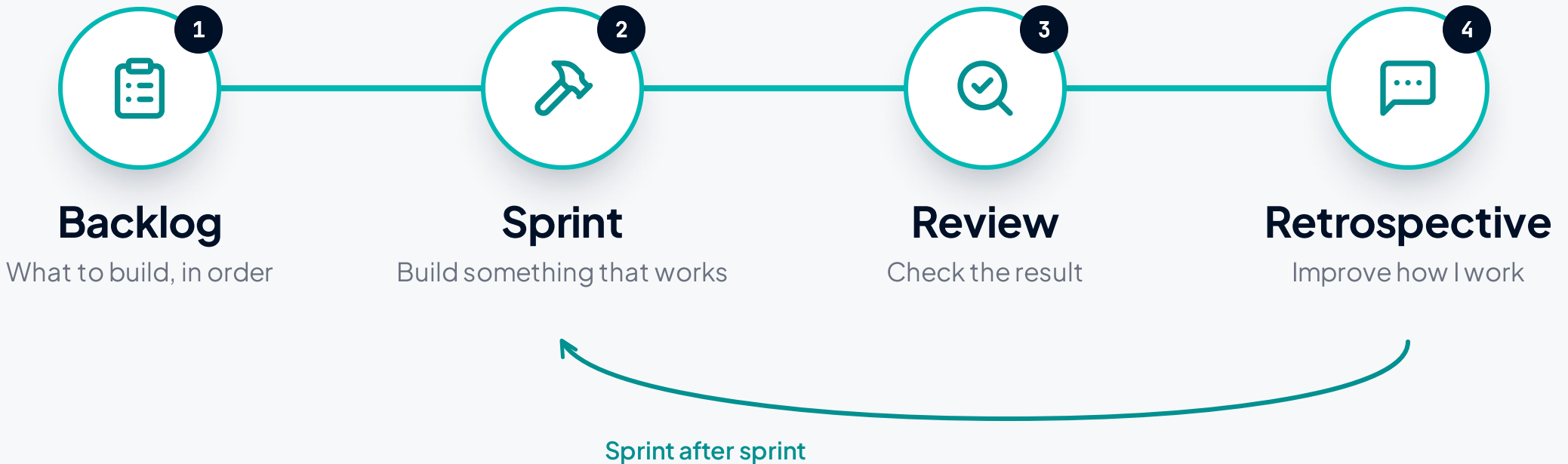


Keep the human in
control



METHODOLOGY

Scrum, adapted to working alone



Tracked in the repository

Epics, tasks, review checklists



Verified automatically

Automated checks on every change



REQUIREMENTS

Functional requirements

Candidate

- ✓ Discover jobs
- ✓ Build a profile from a CV
- ✓ Apply in a few clicks
- ✓ Follow every application

Recruiter

- ✓ Create and publish jobs
- ✓ Manage candidates
- ✓ Run the hiring pipeline
- ✓ Review evidence
- ✓ Schedule interviews
- ✓ Use AI assistance

Platform

- ✓ Multi-tenant organizations
- ✓ Roles and access
- ✓ Billing and entitlements
- ✓ Protected files
- ✓ Notifications

PLATFORM

Each company is an isolated organization, with roles, plans and entitlements, private files and notifications.



REQUIREMENTS

Non-functional requirements



Security

A CV is personal data



Reliability

An application must survive a failure



Scalability

Room to grow with usage



Explainability

AI results must be justified

EXPLAINABILITY

Every result is linked to evidence from the CV, and scored by clear rules.



— SOLUTION

PEAXIS: two experiences, one platform



PEAXIS Jobs

For candidates

Discover jobs · profile from CV · apply · track



PEAXIS Hire

For hiring teams

Jobs · pipeline · evidence · interviews · AI assistance

SHARED PLATFORM

Identity

Organizations

Permissions

Files

Billing

AI



SOLUTION

PEAXIS Jobs: the candidate experience

AB Following Amine

01 Discover

02 Job details

03 Apply with CV

04 Track application

peaxis.com/jobs

My applications

N

Backend Engineer
Northwind · Applied today

Under review

✓

Applied

2

Screening

3

Interview

4

Offer

L

DevOps Engineer
Lumen Labs · Applied 5 days ago

Interview



SOLUTION

PEAXIS Hire: the recruiter workspace

AB Following Amine

northwind.peaxis.com/pipeline

Backend Engineer

Pipeline · 6 candidates

KanbanList

Applied2

KH

Karim Haddad
Full-stack developer

Alignment **62%**

OC

Omar Chaabane
Backend developer

Alignment **71%**

Screening1

SM

Sarah Mansour
DevOps engineer

Alignment **78%** ✓ Reviewed

Interview2

LG

Lina Gharbi
Backend engineer

Alignment **90%** ✓ Reviewed

AB

Amine Benali
Backend engineer · 4 yrs

Alignment **85%**

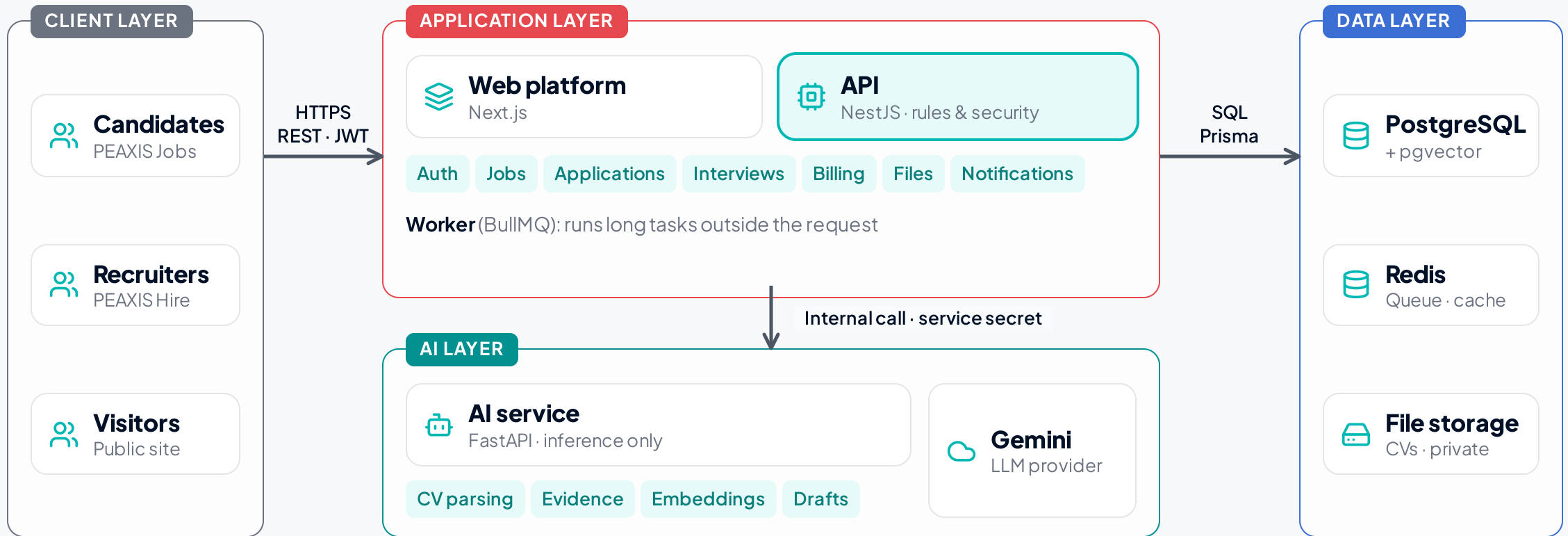
📅 Interview · Thu 10:00

Offer0



— ARCHITECTURE

Logical architecture

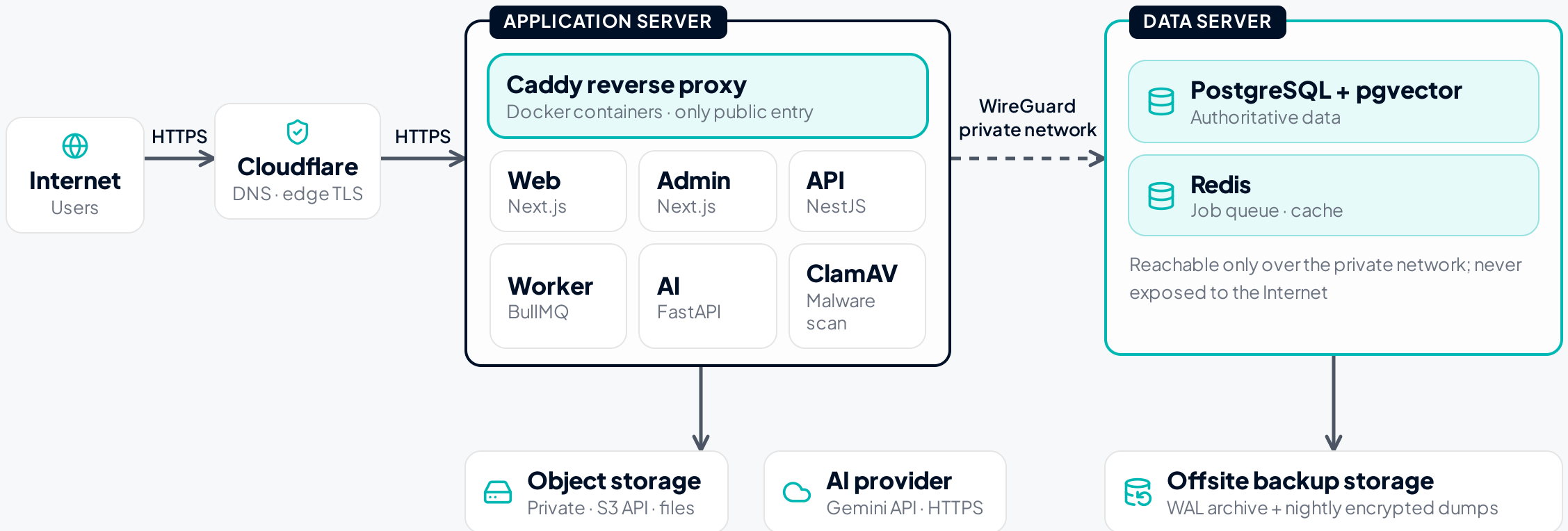


The API is the single authority: the AI layer never touches the data layer.



— ARCHITECTURE

Physical architecture



The public application and the data layer are separated; the database and cache are never exposed to the Internet.



AI

How PEAXIS understands a CV

AMINE BENALI

Backend Engineer · Tunis

EXPERIENCE

Backend Engineer · Northwind, 2021 – 2025

- Built REST APIs with **NestJS** serving 40k users
- Designed **PostgreSQL** schemas and migrations
- Containerized services with **Docker**

SKILLS

TypeScript, NestJS, PostgreSQL, Docker, CI/CD

EDUCATION

Software Engineering degree, 2021



Candidate profile

✓ Checked against source

ROLE	Backend Engineer	✓
EXPERIENCE	Northwind · 2021 – 2025 (4 years)	✓
SKILLS	NestJS PostgreSQL Docker TypeScript	✓
EDUCATION	Software Engineering · 2021	✓

Extract content



AI structures information



Validate against source



Profile & evidence

PEAXIS does not blindly trust generated output. Parsed information is checked against the source before it becomes usable evidence.



AI

Why 85%?

JOB REQUIREMENTS

NestJS Required

PostgreSQL Required

Docker Required

3+ years backend Required

Kubernetes Nice to have

CANDIDATE EVIDENCE

→ *"Built REST APIs with NestJS..."*

→ *"Designed PostgreSQL schemas..."*

→ *"Containerized services with Docker..."*

→ *"Backend engineer, 2021 - 2025"*

→ *No evidence found in the CV*

ASSESSMENT

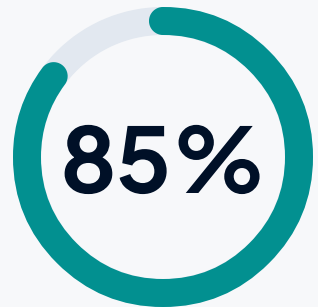
→ ✓ Satisfied

→ ✓ Satisfied

→ ✓ Satisfied

→ ✓ Satisfied

→ ⓘ Needs verification



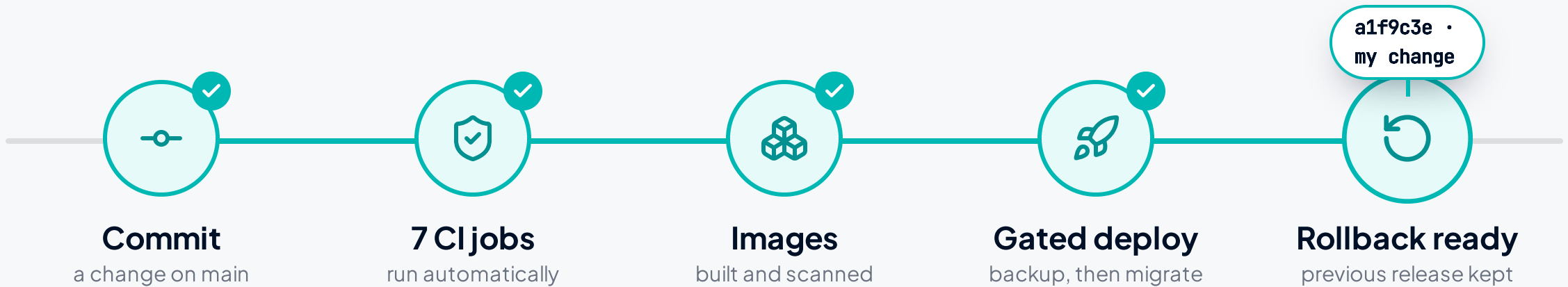
ALIGNMENT

Every result can be traced back to evidence.

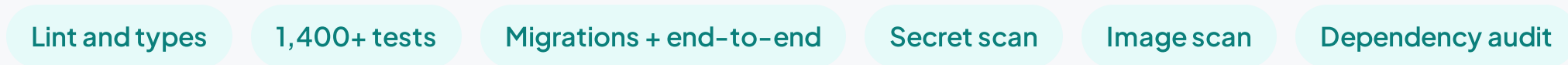


PRODUCTION

From commit to production, safely



WHAT THE CI JOBS CHECK





PRODUCTION

We can see what the system is doing

COLLECTED FROM

Application server
Data server

node exporter · cAdvisor · Caddy ·
PostgreSQL & Redis exporters



Grafana Alloy

One agent per server



Grafana Cloud

 Prometheus

 Loki logs

 Dashboards

 Alerts

Plus synthetic checks on the public site and API.



Servers & containers

CPU, memory, disk, network



API & AI service

Errors, latency, provider calls



Queue & worker

Backlog, failures, heartbeat



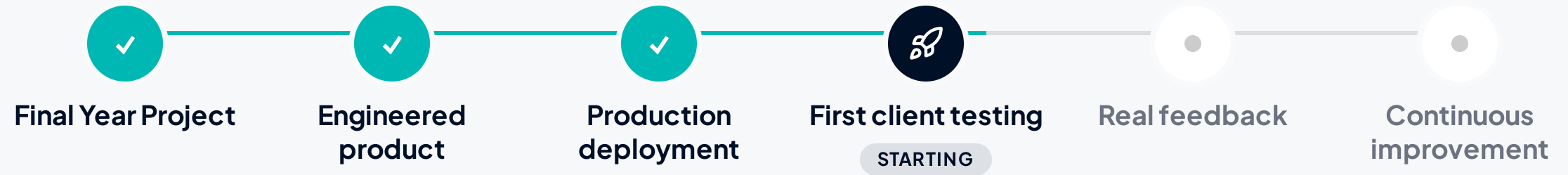
Data & backups

PostgreSQL, Redis, backup status



— PRODUCTION

From Final Year Project to **real product**



PEAXIS was not built only to pass a school project. It is a SaaS product for real companies, and the first ones are starting to test it.



PERSPECTIVES

Where PEAXIS goes next

DIRECTION 1

Deepen PEAXIS Hire

Collaboration

Scorecards, feedback, shared ownership

Communication hub

Candidate email and invitations in one place

Source-linked copilot

Comparisons, interview kits, evidence gaps

Enterprise readiness

Fine-grained permissions, integrations, analytics



DIRECTION 2 FUTURE

PEAXIS HR

Beyond hiring: the employee lifecycle. Possible areas, not commitments:

Employee records

Leave & attendance

Documents

Performance

Employee lifecycle



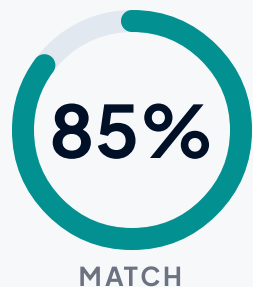
— LIVE DEMO —

PEAXIS in action



CONCLUSION

Back to our first question: would you invite a candidate with an 85% match?



Now we have something better than a number: **the evidence behind it.**

NestJS

✓ Satisfied

PostgreSQL

✓ Satisfied

Docker

✓ Satisfied

3+ years backend

✓ Satisfied

Kubernetes

⚠ Needs verification

THAT IS WHAT I BUILT WITH PEAXIS



One recruitment platform

Jobs for candidates, Hire for recruiters



Explainable AI

Every result linked to evidence



Production engineering

Built to run for real users

Show the evidence. **Let the human decide.**



— THANK YOU

Questions?

Thank you to the jury, and to everyone here, for your attention.

BIBANI Mohamed Habib Allah

PEAXIS · Final Year Project · ESPRIT 2026

Show the evidence. Let the human decide.